

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	30% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2024	Date:	23-08-2026

Code of Conduct

I Aswanth NN (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Aswanth NN

Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words.	Q1
Syntax-Driven Semantic Analysis	Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.	Q3
Representing Linguistically Relevant Concepts	Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.	Q4
Information Retrieval & Syntax-Driven Semantic Analysis	Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.	Q5

Annexure A – Comparative Analysis

1. A conversational AI system must identify grammatical relationships in user queries containing long-distance dependencies. The developers are comparing constituency parsing and dependency parsing.

Analyze how the two representations differ in capturing hierarchical and word-to-word relationships, and justify which approach is more suitable for extracting meaningful relationships from conversational text.

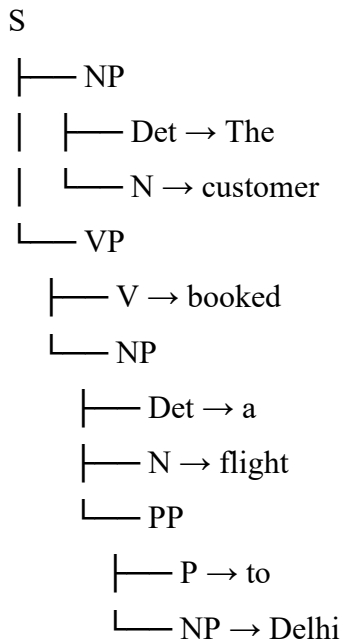
Answer:

Constituency parsing represents a sentence as a hierarchy of phrases or constituents. It divides a sentence into structures such as noun phrases (NP), verb phrases (VP), and prepositional phrases (PP).

For example:

“The customer booked a flight to Delhi.”

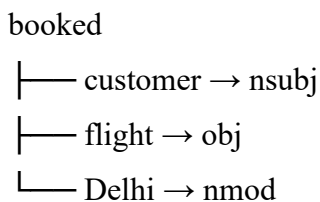
A constituency structure can be represented as:



It clearly shows the hierarchical structure of the sentence.

Dependency parsing, on the other hand, represents direct relationships between individual words. One word acts as the head and other words depend on it.

For the same sentence:



Dependency parsing directly shows which words are grammatically related.

Comparison:

Feature	Constituency Parsing	Dependency Parsing
Main representation	Phrase hierarchy	Word-to-word relationships
Focus	Constituents and phrases	Head-dependent relationships
Structure	Tree of phrases	Dependency tree
Long-distance relationships	Can represent them	Directly represents them
Semantic relationship extraction	Moderate	Strong
Conversational text	Useful	Highly useful

For conversational AI, **dependency parsing is generally more suitable for extracting meaningful relationships** because it directly identifies relationships between words such as subject, object, modifier, and destination.

For example:

customer → booked → flight

flight → to → Delhi

This structure can be directly converted into information needed for tasks such as intent detection and information extraction.

Therefore, **constituency parsing is better for understanding hierarchical phrase structure, while dependency parsing is generally better for extracting word-level grammatical and semantic relationships from conversational text.**

2. An NLP system processes partially typed search queries such as “best hotels near...” and “book a flight from Chennai to...”. The system currently uses conventional top-down parsing, while the developers are considering Earley parsing.

Analyze how the two approaches behave when input is incomplete or ambiguous, and justify which parsing strategy is more appropriate for interactive NLP applications.

Answer:

Top-down parsing starts with the start symbol of the grammar and attempts to generate the input sentence by selecting grammar rules.

For example:

“Book a flight from Chennai to...”

The parser may attempt to construct:

$S \rightarrow VP$

$VP \rightarrow V \text{ NP PP PP}$

However, the input is incomplete because the destination has not been provided.

A top-down parser may fail to construct a complete parse or may need to wait for additional input.

It can also experience **backtracking** when multiple grammar rules are possible.

Earley parsing is a chart-based parsing technique that can maintain partial parsing states.

For:

“Book a flight from Chennai to...”

it can maintain a partial structure such as:

Book → Action

flight → Object

Chennai → Source

to → Destination marker

The parser can wait for the missing destination instead of completely failing.

Earley parsing also handles ambiguous input by maintaining multiple possible interpretations.

Comparison:

Feature	Top-Down Parsing	Earley Parsing
Incomplete input	Difficult	Handles well
Ambiguity	May require backtracking	Maintains alternatives
Backtracking	Can be high	Reduced through chart
Partial parsing	Limited	Good
Interactive systems	Less suitable	Highly suitable
Long/complex input	Can become inefficient	Better suited

For interactive NLP applications, **Earley parsing is more appropriate** because users frequently type incomplete queries and expect the system to provide suggestions while they are still typing.

For example:

“best hotels near...”

can be processed as a partial query, allowing the system to predict possible locations.

Therefore, **Earley parsing provides better support for incomplete and ambiguous input and is more suitable for interactive search and conversational applications.**

3. A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. The developers are comparing CFG, PCFG, and neural constituency parsers.

Analyze how each approach represents and resolves syntactic ambiguity, and justify which approach would provide the best balance between interpretability and practical accuracy.

Answer:

CFG

A **Context-Free Grammar (CFG)** represents syntactic structures using grammar rules.

For example:

$S \rightarrow NP VP$

$VP \rightarrow V NP$

$NP \rightarrow Det N$

If a sentence has multiple valid structures, CFG can generate multiple parse trees.

However, CFG does not provide probabilities to determine which parse is more likely.

Therefore:

CFG $\rightarrow$ Multiple possible parses

but it does not automatically select the best one.

PCFG

A **Probabilistic Context-Free Grammar (PCFG)** extends CFG by assigning probabilities to grammar rules.

For example:

$VP \rightarrow V NP$ [0.70]

$VP \rightarrow VP PP$ [0.30]

If multiple parses are possible, the probabilities can be used to select the most likely parse.

Therefore:

CFG + probabilities $\rightarrow$ Most probable parse

PCFG is more effective than basic CFG for structural ambiguity while remaining relatively interpretable.

Neural Constituency Parser

A neural constituency parser uses machine-learning models, often based on neural networks, to learn syntactic structures from large datasets.

It can capture complex linguistic patterns that may be difficult to represent manually using CFG rules.

Advantages include:

- high practical accuracy
- ability to learn complex patterns
- better handling of natural language variation
- less dependence on manually written grammar rules

However, neural models are generally less interpretable than CFG and PCFG.

Comparison:

Feature	CFG	PCFG	Neural Parser
Grammar rules	Explicit	Explicit	Learned
Ambiguity handling	Generates alternatives	Ranks alternatives	Learns likely structure
Probability	No	Yes	Yes
Interpretability	High	High	Lower
Practical accuracy	Moderate	Better	Generally high
Data requirement	Low	Moderate	High

For a grammar-checking system, **PCFG provides the best balance between interpretability and practical accuracy** when the goal is to understand why one syntactic interpretation was selected.

A neural parser may provide higher accuracy in large-scale real-world applications, but PCFG is easier to explain and debug.

Therefore:

CFG is best for simple and transparent grammatical rules, PCFG is best for interpretable ambiguity resolution, and neural constituency parsing is best when maximum practical accuracy is the primary goal.

4. A multilingual NLP system must process sentences in which grammatical information depends on number, gender, tense, and person. The developers are considering feature structures and traditional context-free grammar rules.

Analyze how feature structures extend basic grammar representations and justify their usefulness for enforcing grammatical constraints in multilingual systems.

Answer:

A traditional CFG represents grammatical structures using rules such as:

$S \rightarrow NP VP$

$NP \rightarrow Det N$

$VP \rightarrow V NP$

However, these rules do not naturally represent detailed grammatical information such as:

- Number
- Gender
- Person
- Tense
- Case
- Agreement

For example, a basic CFG may allow:

“The students is reading.”

because the grammar only checks whether the structure is:

NP + VP

It does not necessarily check that **students** is plural and **is** is singular.

Feature Structures

Feature structures attach grammatical properties to linguistic elements.

For example:

NP

Number = Plural

Person = Third

and:

VP

Number = Plural

Person = Third

Tense = Present

The system can require the features to agree.

For example:

NP:

Number = Plural

VP:

Number = Plural

This allows:

“The students are reading.”

but rejects:

“The students is reading.”

Feature structures can also represent gender:

Gender = Masculine

or:

Gender = Feminine

and tense:

Tense = Past

Comparison:

Feature	Basic CFG	Feature Structures
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Phrase structure	Yes	Yes
Number	Limited	Yes
Gender	Limited	Yes
Person	Limited	Yes
Tense	Limited	Yes
Agreement	Difficult	Effective
Multilingual use	Limited	Strong

Feature structures are particularly useful in multilingual NLP because different languages may have different agreement requirements.

They allow grammatical constraints to be explicitly represented and checked during parsing.

Therefore, **feature structures provide a more expressive representation than basic CFG and are highly useful for multilingual grammar processing, agreement checking, machine translation, and syntactic analysis.**

5. A voice assistant must interpret commands containing verbs with different argument requirements, such as “give the book to John,” “sleep,” and “put the file on the desk.” The developers are considering subcategorization frames.

Analyze how subcategorization information helps determine valid sentence structures and justify its importance in semantic and syntactic analysis.

Answer:

Subcategorization describes the types and number of arguments that a verb requires to form a grammatically complete sentence.

Different verbs require different structures.

Example 1: “Give the book to John.”

The verb **give** normally requires:

Subject + Verb + Object + Recipient

Its subcategorization frame can be represented as:

give → NP Subject + NP Object + PP Recipient

Therefore:

Someone → gives → book → to John

The system can identify:

Agent = Someone

Action = Give

Object = Book

Recipient = John

Example 2: “Sleep.”

The verb **sleep** does not require an object.

Its frame is:

sleep $\rightarrow$ NP Subject

Therefore:

“John sleeps.”

is complete.

A sentence such as:

“John sleeps the book.”

is normally invalid because **sleep** does not require or accept a direct object in this sense.

Example 3: “Put the file on the desk.”

The verb **put** requires both an object and a location.

Its frame is:

put $\rightarrow$ NP Subject + NP Object + PP Location

Therefore:

Agent = Someone

Action = Put

Object = File

Location = Desk

The phrase:

“John put the file.”

is incomplete in standard usage because **put** normally requires a destination or location.

Comparison:

[illegible]

Give Subject + Object + Recipient Give the book to John

Sleep Subject	John sleeps
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Put Subject + Object + Location Put the file on the desk

Subcategorization is important because it helps the NLP system determine:

1. **Which arguments a verb requires**
2. **Which sentence structures are grammatically valid**
3. **Which words function as semantic roles**
4. **Whether an input command is complete**
5. **How to convert natural language into structured commands**

For a voice assistant, this is particularly useful because the system can convert commands into structured representations.

For example:

"Give the book to John"

becomes:

Action = Give

Object = Book

Recipient = John

while:

"Put the file on the desk"

becomes:

Action = Put

Object = File

Location = Desk

Therefore, **subcategorization frames connect syntactic structure with semantic roles and are essential for reliable command understanding in conversational and voice-based NLP systems.**

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Scope of Items	Comparison Depth	Use of Eviden ce	Critical Thinking	Clarity of Present ation	Sub. Total	Total %
1	4	3	4	4	4	19	93
2	4	3	4	4	4	19	
3	4	3	4	4	4	19	
4	4	3	4	4	4	19	
5	4	3	4	4	4	19	

Faculty In-charge	Course Coordinator	Program Director
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Date: _____	Date: _____	Date: _____